C-V2X Monitoring and Anomaly Detection

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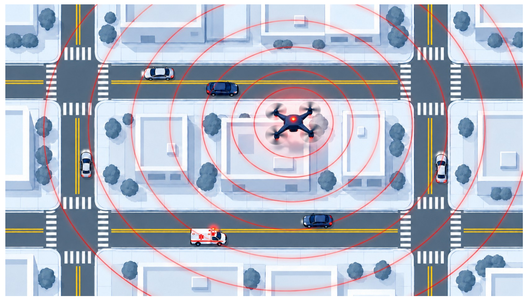
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Abstract and motivation

False reports can divert responders, while flooding can suppress safety messages. Message-only monitoring offers limited insight into radio activity. We link C-V2X messages to radio evidence and demonstrate flooding detection in a controlled experiment. Regional correlation and camera-assisted verification remain proposed.



False report (illustrative)



RF jamming (illustrative)

Roadside prototype



USRP B210 receiver
Cohda MK6 OBUs
5.915 GHz, 20 MHz channel
30.72 MS/s sampling
10-Hz BSMs with retransmissions
Core Ultra 9 host, 64 GB RAM

Reception validation

99.49% stationary BSM decoding
~20 ms mean receive-to-BSM latency
>1 km with favorable line of sight

Stationary: 1,750 / 1,759 BSMs at ~20 m.
Moving range depended on route visibility.

Support: NSF CNS-2321531, CNS-2323174, DMS-2436333, ITE-2453815; USDOT CCAT; LRRB/MnDOT 1058817.

Monitoring system

Raw I/Q samples → SCI + transport block decoding → RF-time observations → Features + anomaly alerts

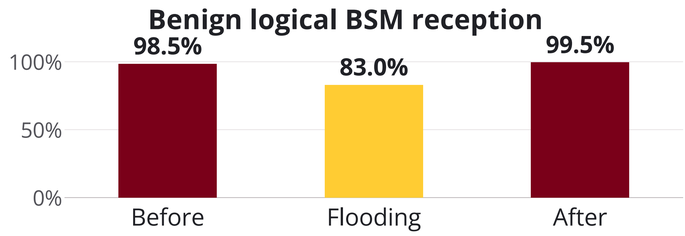
A passive roadside unit (PRU) decodes sidelink control information (SCI) and Basic Safety Messages (BSMs). RF-time records link bursts, receiver settings, failed attempts and retransmissions.
Channel analyzers derive RF, occupancy and reception features and identifier-linked tracks. Independent detectors apply thresholds and temporal rules, with alerts linked to supporting records. New analyzers reuse the evidence without decoder changes.

Regional extension (proposed)

PRUs would send compact reports to a regional edge. CNN-based RF positions and BSM reports would be compared with calibrated camera tracks to check consistency and screen training labels. Regional correlation and camera-assisted localization remain to be evaluated.

Message-flooding attack and detection

Two benign OBUs operated during a 60-s trial: 10 s baseline, 40 s flooding and 10 s recovery. A B210 flooder replayed a fixed waveform without sensing-based resource selection.



0.64% to 32.24%

decoded-resource occupancy

Alert at t = 11 s
Flooding began at t = 10 s

333 active subframes/s

Five Layer-2 IDs shared one BSM temporary ID. Exact identifier matches grouped them into one track. RF/timing similarity provided a separate consistency check.
Each 1-s window combines normalized load (70%), identity multiplicity (15%) and benign impact (15%). Eligible sources can trigger at scores ≥ 0.70. A severe-load rule can trigger independently.
The detector flagged the flooder and neither benign OBU.